

GOKULNAATH GOVINDARAJ

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SUMMARY

Full-stack engineer with production experience building Java REST APIs and Python microservices, including a FastAPI service where diagnosing a 2,000 RPS PostgreSQL bottleneck cut p95 latency from 1.8s to 200ms. Experienced with RabbitMQ async pipelines across 8 concurrent workers, hybrid BM25 and dense retrieval against a Weaviate vector store, and React frontends, with RAG architecture and LLM evaluation built into production document workflows.

SKILLS

AI & LLM Systems: LLM API integration (Claude, Gemini 2.5 Flash via google-genai, GPT-5.1), RAG pipelines, hybrid retrieval (BM25 + dense + RRF), ChromaDB, Weaviate, structured output / JSON-schema enforcement, OCR ingestion (Docling, PyMuPDF), prompt engineering, LLM evaluation frameworks

Languages: Python, Java (Spring Boot, Android), JavaScript / TypeScript, SQL, Go (developing)

Backend & APIs: FastAPI (async, Pydantic validation), Flask, RESTful API design, GraphQL, SQLAlchemy (ORM, Alembic), Spring Boot, RabbitMQ (async job queues, concurrent workers), microservices architecture, Gunicorn

Cloud & Infrastructure: Docker, GCP (Cloud Run, GKE, Vertex AI, Cloud Logging/Monitoring), AWS (RDS, EC2, S3), PostgreSQL (schema design, indexing, connection pooling), Kubernetes, GitHub Actions, GitLab CI/CD

EXPERIENCE

- AI Trainer (Software Engineering)** - Handshake AI Apr 2026 - Present | Remote - Contract
- Drive multi-agent software engineering workflows on Handshake AI's Helix project, designing Ralph loop autonomy patterns and atomic prompt scaffolding that guide Claude Code agents to solve complex tasks without implementation leakage.
 - Deliver Go-based backend solutions with high-fidelity sqlmock test coverage as part of software engineering bounty work, producing 37+ verified pull requests with ~89% automated gate pass rate across containerized Kubernetes environments.
- Software Engineer** - AriesView Aug 2025 - Present | Boston, MA
- Architected the RAG pipeline for a CRE due diligence platform, designing chunking strategies, embedding pipelines, and hybrid BM25 + dense retrieval against a Weaviate vector store to power document-aware Q&A over financial and lease documents.
 - Engineered ingestion pipelines using Docling and PyMuPDF for OCR-based document processing, covering structured text extraction and document-aware chunking for downstream retrieval.
 - Diagnosed PostgreSQL connection pool saturation under ~2,000 RPS using k6 load tests, reducing p95 API latency from 1.8s to 200ms and cutting timeout errors from 1.2% to 0.1% by right-sizing Gunicorn workers on an 8-vCPU GCP deployment.
 - Designed a RabbitMQ job queue distributing asynchronous OCR and LLM extraction across 8 concurrent workers, parallelizing multi-file ingestion with metadata-driven job state tracking and durable per-job retry semantics.
 - Developed FastAPI endpoints with SQLAlchemy ORM models, Pydantic request validation, and Alembic migrations to manage schema evolution across the production PostgreSQL database.
- Programming Intern** - Porter Lee Corporation May - Aug 2024 | Schaumburg, IL
- Navigated a 30,000+ line Java Android codebase to build REST API endpoints for an evidence management system, reducing JSON deserialization time by ~180ms per request through optimized parsing logic.
 - Implemented JWT-based authentication securing API access across all endpoints and introduced multithreaded background sync that reduced average UI freeze duration by ~300ms during data refresh cycles.
- Programmer Analyst Trainee** - Cognizant Sep 2022 - Mar 2023 | Chennai, India
- Participated in Agile ceremonies including sprint planning, stand-ups, and retrospectives, collaborating with developers to validate features before release.

PROJECTS

- RAG System - Financial Document Intelligence** ChromaDB · BGE-small · Gemini 2.5 Flash · FastAPI · BM25 + RRF
- Enforced a strict layered pipeline (Extractor, Chunker, Embedder, Indexer, Retriever, Answerer) with single-responsibility contracts and per-phase validation gates that drop bad data at the boundary; loud token-truncation warnings surfaced an upstream chunker gap that silent truncation would have hidden.
- AccioFirst - Ghost-Job Intelligence Platform** Next.js · FastAPI · Gemini 2.5 Flash-Lite · BrightData · Supabase
- Built a multi-signal job intelligence scoring pipeline in FastAPI that ingests a job URL, coordinates BrightData scraping, career-page cross-referencing, and company hiring-signal retrieval, then feeds structured output to Gemini 2.5 Flash-Lite to return a scored ghost-job verdict in under 5 seconds.

EDUCATION

- Master of Science in Computer Science** | Wright State University Aug 2023 - Dec 2025
- Honors and Recognition:**
- 1. Runner-Up, Wright Venture 2026 - AccioFirst - Wright State University - April 2026 \$2,000 funding awarded
 - 2. Technical Lead, NSF I-Corps Cohort - University of Connecticut - September 2024 \$1,000 funding awarded